

P100 Service: teensycfg (display and button customization)

Summary

Some elements of the display and some of the buttons of the p100 are controlled by the Teensy microcontroller.

This customizations are managed by *teensycfg* service.

Udev Rules

The following udev rules must be defined on `/etc/udev/rules.d/49-teensy.rules`:

```
1 ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04[789B]?", ENV{ID_MM_DEVICE_IGNORE}="1",
  ENV{ID_MM_PORT_IGNORE}="1"
2 ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04[789A]?", ENV{MTP_NO_PROBE}="1"
```

Configuration

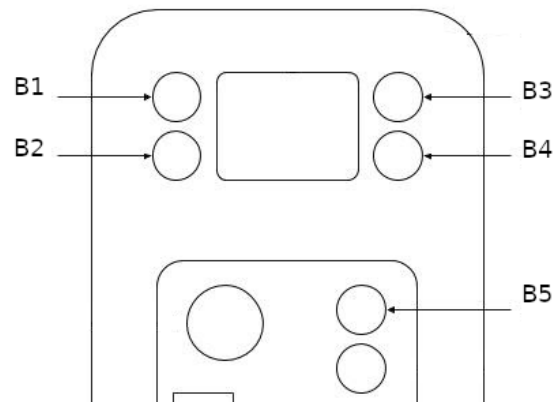
The configuration file is located at `/etc/teensyconfig` as a *json* file:

```
1 {
2   "buttons": {
3     "1": [
4       "inop"
5     ],
6     "2": [
7       "inop"
8     ],
9     "3": [
10      "internet"
11    ],
12    "4": [
13      "inop"
14    ],
15    "5": [
16      "wifi_on",
17      "wifi_off"
18    ]
19  },
20  "callbacks": {
21    "wifi_off": "led5off",
22    "wifi_on": "led5on"
23  },
24  "wap": "on",
25  "title": "Ethiopian Airlines"
26 }
```

Buttons

for the buttons, we can assign a list of consecutive actions. In the example above, button 5 has been configured so that the Wi-Fi is turned on the first time the button is pressed, and turned off the next time the button is pressed, always at the same order and at the end of the last action it starts again.

- `inop` (button will not perform any action)
- `wifi_on` (button will turn on the Wi-Fi)
- `wifi_off` (button will turn off the Wi-Fi)
- `internet` (show internet status on display)



callbacks

Some actions may trigger callbacks. `wifi_on` and `wifi_off` can trigger:

- `led1on` (Turn on the light around button 1)
- `led1off` (Turn off the light around button 1)
- `led2on` (Turn on the light around button 2)
- `led2off` (Turn off the light around button 2)
- `led3on` (Turn on the light around button 3)
- `led3off` (Turn off the light around button 3)
- `led4on` (Turn on the light around button 4)
- `led4off` (Turn off the light around button 4)

wap

You can configure the initial status of wap:

- `on`
- `off`

title

You can define which title you want to appear at the bottom of the screen.

Execution

The *teensycfg* service is managed by **upstart** (Ubuntu 14.04).

You can control this service as usual:

```
1 service teensycfg status
2 service teensycfg start
3 service teensycfg stop
4 service teensycfg restart
```